

Detection Filter History — ATS Triton Receiver Interface

Applies to: Triton Receiver Interface v1.16 and later

Last updated: October 2026

This document describes the post-processing filters in the Triton Receiver Interface and records how they have changed from version to version.

If you are reprocessing older data, or comparing results produced by two different versions of the software, use the version table in Part 2 to see whether a filter changed in between.

Part 1 — What each filter does

The filters are reached through **File** → **Filter Files** (Ctrl+F), which opens the **Filter Options** dialog. Every option in that dialog has a tooltip: hover your mouse over a checkbox or field to see what it does.

The dialog is divided into two groups.

Filter Methods — choose which detections to keep

Filter	Default	What it does
False Positive	On	Removes multipath echoes and spurious detections by checking the interval between detections against the tag's pulse repetition interval (PRI). A detection is kept when it falls inside the expected PRI window.
Code	Off	Keeps only detections whose tag code appears in the code list file you select. Use this to pull your own study tags out of a file that also contains other projects' tags.
Press Sensor Decoding	Off	Decodes pressure (depth) sensor tags and keeps only readings inside the depth range you specify. Cannot be combined with Temp/Press Sensor Decoding — the two target different tag types.
Temp/Press Sensor Decoding	Off	Decodes three-pulse tags that report both temperature and pressure, and keeps only readings inside both the temperature and the depth range you specify.

Data Quality / Verification — check the data file

These options inspect the file rather than deciding which fish detections to keep. They are useful when a receiver's clock is suspect.

Check	Default	What it does
Verify Tag Times	Off	Diagnostic only. Cross-checks each timestamp against the receiver's internal DSP counter and adds an <code>Adjusted_Date_Time</code> column alongside the original, so you can compare the two. Your data is not altered.
Correct Tag Times	Off	Replaces the <code>DateTime</code> column with the reconstructed timestamp before the other filters run. Requires Verify Tag Times to be enabled. Use this when a deployment's clock is known to be wrong.
Flag Out-of-Order Rows	Off	Reports rows where the receiver's internal counter runs backwards, which indicates detections logged out of chronological order. GPS rows are excluded from this check.

Output options

The same dialog controls which data columns are carried into the filtered output (tilt, pressure, temperature, voltage, signal strength and others), and whether a summary file and pressure graph are produced alongside the filtered CSV.

Filtered output is written to the `filter_output` folder unless you choose another location.

Supported input files

Filters accept both file types the system produces:

- Detection logs downloaded from the receiver's SD card.
- Real-time logs recorded by the interface during a live session (`..._dat.csv`).

Part 2 — What changed, by version

A note on version numbers. Releases use a two-digit minor number: v1.16 is "one point sixteen", v1.28 is "one point twenty-eight". There has never been a v1.6 or v1.06.

v1.16 — October 2025

First release with filtering.

- Added the **Code** filter (filter a detection file against a tag code list).

- Added the **False Positive** filter with a configurable PRI window.
- Added the default `filter_output` folder for filtered results.
- Pressure sensor tag decoding started but was not yet usable.

v1.23 — February 2026

- **Press Sensor Decoding finished** and usable on real acoustically coded pressure tags.
- **Added time verification.** Receiver timestamps could occasionally be logged out of sequence or at implausible values; this check reconstructs the expected time from the receiver's internal counter so the two can be compared.
- Time verification can be run on its own, without also applying the Code or False Positive filters.

v1.25 — May 2026

- **Filtering extended to real-time logs.** Logs recorded live by the interface (`..._dat.csv`) can now be filtered, not only files downloaded from the SD card.
- Added **Flag Out-of-Order Rows**.
- Added **Save Filtered Detections**, which writes a second live CSV during a real-time session containing only detections matching your active filter.

v1.27 — June–July 2026

A correctness release. Several of these were genuine data errors, not cosmetic changes. **If you produced corrected timestamps with v1.23–v1.26, reprocess those files with v1.27 or later.**

- **Files with more than one time marker were reconstructed incorrectly.** In a file containing several receiver time markers, the software could attach every detection to a single reference time, placing reconstructed timestamps days to months away from the true time. Files with one marker were unaffected, which is why this went unnoticed. Timestamps are now anchored to the correct marker.
- **Receiver restarts were mistaken for counter rollovers.** The internal counter can run backwards for two reasons: it wrapped around its maximum, or the receiver restarted it. Treating a restart as a wrap added a spurious jump of roughly eleven days. The two cases are now told apart.
- **Pulling the SD card mid-deployment shifted later data backwards.** Data recorded after a log resume could be pinned to a stale reference time — in one case data recorded on 28 January was reported near 8 December. Fixed.
- **Corrected dates could print with the year 0025.** A two-digit year was re-read as a four-digit year in the output column. Filtering decisions were unaffected; only the printed date was wrong.

- **Filter Options dialog regrouped and relabeled.** The checkboxes were split into "Filter Methods" and "Data Quality / Verification" so it is clear which options discard detections and which only inspect the file. Wording was made tag-centric. Behavior was unchanged by the renaming.

Old label (v1.26 and earlier)	New label (v1.27 and later)
Apply Filter	False Positive
Apply Code Filter	Code
Apply Press Sensor Filter	Press Sensor Decoding
Verify Detection Times	Verify Tag Times
Run Through filters after verifying	Correct Tag Times

- Added the **About** button in the Filter Options dialog.

v1.28 — August 2026

No changes to filter behavior. Fixes and label cleanup elsewhere in the software.

Questions

Contact Advanced Telemetry Systems — <https://atstrack.com>

Advanced Telemetry Systems, Isanti, Minnesota.